

Dr. Babasaheb Ambedkar Technological University's
Institute of Petrochemical Engineering, Lonere –Raigad 402 103
Summer Examination – May 2016

Course : Electrical Engineering
Subject: Applied Mathematics IV

Date: 03 May 2016

Sem.IV
Marks: 50

Instructions to the Students:

- 1) Attempt any **Five Questions** from Question No.1 to Question No.6.
- 2) If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1) a) Express $\sqrt{3}\left(\cos\frac{8\pi}{6} + i\sin\frac{8\pi}{6}\right)$ in the $x + iy$ form. (5)

Prove that $\left[\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right]^8 = \cos(4\pi-8\theta) + i\sin(4\pi-8\theta)$. (5)

Q.2) Solve any **TWO** of the following. (10)

- a) Find all values of $(1+i)^{1/3}$.
- b) Express $\frac{1+2i}{1-3i}$ into exponential form.
- c) Find the complex cube roots of unity.

Q.3) Solve the following differential equations: (**ANY TWO**) (10)

a) $\tan y \frac{dy}{dx} = \sin(x+y) - \sin(x-y)$, b) $(x^2 + y^2 - a^2)xdx + (x^2 - y^2 - b^2)ydy = 0$, c) $\frac{dy}{dx} + y \tan x = \cos^2 x$.

Q.4) a) Define differential equation and state the order and degree of $y + \frac{dy}{dx} = \frac{1}{4} \int y dx$. (5)

b) A resistance of 100 ohm, an inductance of 0.5 henry are connected in series with a battery of 20 volts. Find the current in the circuit at $t = 0.5$ sec. if $i = 0$ at $t = 0$. (5)

Q.5) a) Find the Laplace Transform of the function defined by $F(t) = \begin{cases} 4, & 0 \leq t \leq 1 \\ 3, & t > 1 \end{cases}$. (5)

b) Find $L^{-1}\left[\frac{1-7s}{(s-3)(s-1)(s+2)}\right]$. (5)

Q.6) a) Find the half range Fourier Cosine Series for the function $f(x) = x$ in $(0, \pi)$. (5)

b) Find the Fourier Transform of $f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$. (5)

ALL THE BEST